Competitive Exam Practice 2 — Answer Key

Number Systems, Representation, and Arithmetic Hardware

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Q1 **(C).** $43 = 32+8+2+1 = 00101011$ in 8-bit binary. Since 43 is non-negative, its 2's complement representation is the plain binary value. (01010101 = 85; 11010101 and 10101011 have the sign bit set, so they are negative.)

Q2 $\mathbf{-2^{n-1} \le N \le 2^{n-1} - 1}$. With n bits in two's complement, the MSB is the sign bit; the most negative value has MSB 1 and all else 0 (-2^{n-1}), and the most positive has MSB 0 and all else 1 ($2^{n-1}-1$). One of the 2^n bit patterns is wasted by reserving it for the single representation of zero.

Q3 **(C).** Squaring both sides: $(224)_r = (13)_r^2$. Expanding: $2r^2 + 2r + 4 = (r+3)^2 = r^2 + 6r + 9$. Rearranging: $r^2 - 4r - 5 = 0$, i.e. $(r-5)(r+1) = 0$. Since $r > 4$ (digit 4 appears), $r = 5$.

Q4 **15 full adders, 1 half adder.** A ripple-carry adder adds bit 0 with no carry-in, so it needs only a half adder (or a full adder with $C_{in} = 0$); every other bit position (1 through 15) needs a full adder to accept the carry from the previous stage. Hence $n - 1 = 15$ full adders and 1 half adder.

Q5 **(C).** $539_{10} = 0x21B$. The 2's complement of a positive number in k bits is $2^k - x$. Using 12 bits: $-539 = 4096 - 539 = 3557_{10} = 0xDE5$. (Option (C) as printed in the book is DE5; the OCR reading "DES" is a 5/S glyph ambiguity.)

Q6 **(B).** Critical path through the carry-lookahead adder:

- **Generate propagate/generate:** $P_i = A_i \oplus B_i$ and $G_i = A_i \cdot B_i$. XOR is two gate levels (two-level AND–OR), so delay = 2.
- **Carry network:** each C_i is a two-level AND–OR expression (sum of products), delay = 2.
- **Sum:** $S_i = P_i \oplus C_i$, another two gate levels, delay = 2.

Total = $2 + 2 + 2 = 6$ time units.